Stat 293: Tutorial 3

Q1: Permutation Testing

This example uses the smoking.csv dataset used in Chapter 2 of the course notes - it is available on Blackboard.

(a) Write and implement R-code to create 1000 re-ordered samples and obtain a p−value using a permutation test.

```
# 1.1a - Read in the data
smo1a <- read.csv("F:\\Lecture Courses\\STAT 293\\...\\ smoking.csv")
str(smo1a)
```

```
## 'data.frame':   24 obs. of  2 variables:
## $ group: int  1 1 1 1 1 1 2 2 2 2 ...
## $ fev1 : num  4.41 4.96 3.5 3.66 4.68 4.11 3.69 3.9 3.82 4.08 ...
```

```
# 1.1b - Constructing the ANOVA Model
smo1b <- aov(fev1 ~ as.factor(group), data=smo1a)
summary(smo1b)
```

```
                    Df      Sum Sq  Mean Sq   F value   Pr(>F)
## as.factor(group)   3      3.689   1.2297    3.623   0.0309 *
## Residuals         20      6.788   0.3394
```

```
## Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
# 1.1c - Extracting the observed test_statistic F_obs
smo1c <- summary(smo1b)[[1]]["as.factor(group)","F value"]
smo1c
```

```
## [1] 3.623135
```

```
# 1.1d - Construct a distribution of similar test statistics from a cohort of 1,000 re-ordered samples.

        # Define number of re-ordered samples to take
        L <- 1000

        # Create a blank vector called smo1d of length L
        smo1d <- rep(NA,L)

        # Set seed to ensure we obtain the same re-ordered samples each time code is run
        set.seed(12345)

        # Do for i = 1,...,L
        for (i in 1:L) {

                # Obtain a re-ordered sample from the smo1a data set
                group.perm <- sample(smo1a$group)

                # Conduct ANOVA on the re-ordered sample
                this.aov <- aov(fev1 ~ as.factor(group.perm), data=smo1a)

                # Extract the calculated F-Statistic
                summary(this.aov)
                F.perm <- summary(this.aov)[[1]]["as.factor(group.perm)","F value"]

                # Add calculated F-statistic to the ith element of the vector smo1d
                smo1d[i] <- F.perm
                }


# 1.1e - Calculate the p-value for the test.
length(smo1d [smo1d >= smo1c] )/ L
```

```
## [1] 0.039
```

(b) Write down the hypotheses tested in part (a). What conclusion do you draw about the hypotheses?

H0: There are no differences in lung function among different types of smokers.
H1: Lung function differs among types of smokers.

P-value = 0.039
=> Some evidence against the null hypothesis

Q2: Permutation Testing

A study to test whether a teaspoon of honey before bed calms a child's cough was published in the Archives of Peadiatrics and Adolescent Medicine (December 2007). In each of three groups of randomly selected children with an upper respiratory tract infection, symptom scores were measured on the first night when no treatment was given and then again on the second night after treatment was given. One group received no treatment (control), the second received honey and the other an over-the-counter cough remedy. The improvement in symptom scores was the variable of interest. The data are available in the dataset 'crdhoneycough.csv' on Blackboard.

(a) Carry out a permutation test on the data by creating re-ordered samples (by reordering the response variable) **i.e. DO NOT use lmPerm**. Use 5000 re-ordered samples. State your hypotheses and any conclusions you draw

```
# 2.1a - Read in the data
hon1a <- read.csv("F:\\Lecture Courses\\STAT 293\\...\\ crdhoneycough.csv")
str(hon1a)
```

```
## 'data.frame':   105 obs. of  2 variables:
## $ treatment: chr  "H" "H" "H" "H" ...
## $ score    : int  12 11 15 11 10 13 10 4 15 16 ...
```

```
# 2.1b - Constructing the ANOVA Model
hon1b <- aov(score ~ treatment, data=hon1a)
summary(hon1b)
```

```
                Df      Sum Sq  Mean Sq   F value  Pr(>F)
## treatment    2       318.5   159.2     17.51    2.9e-07 ***
## Residuals    102     927.7   9.1

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 2.1c - Extracting the observed test_statistic F_obs
hon1c <- summary(hon1b)[[1]]["treatment","F value"]
hon1c
```

```
## [1] 17.50957
```

```
# 2.1d - Construct a distribution of similar test statistics from a cohort of 5,000 re-ordered samples.

        # Define number of re-ordered samples to take
        L <- 5000

        # Create a blank vector called hon1d of length L
        hon1d <- rep(NA,L)

        # Set seed to ensure we obtain the same re-ordered samples each time code is run
        set.seed(12345)

        # Do for i = 1,...,L
        for (i in 1:L) {

                # Obtain a re-ordered sample from the hon1a data set
                reordered_sample <- sample(hon1a$treatment)

                # Conduct ANOVA on the re-ordered sample
                reordered_perm <- aov(score ~ reordered_sample, data=hon1a)

                # Extract the calculated F-Statistic
                summary(reordered_perm)
                F_extract <- summary(reordered_perm)[[1]]["reordered_sample","F value"]

                # Add calculated F-statistic to the ith element of the vector hon11d
                hon1d[i] <- F_extract
                }


# 1.1e - Calculate the p-value for the test.
length( hon1d[hon1d >= hon1c] )/ L
```

```
# [1] 0
```

H0: There are no differences in symptom score improvements among the three treatments.
H1: Symptom score improvements differ among the three treatments.
P-value = 0
=> Very strong evidence against the null hypothesis

Q3: Two Way ANOVA with lmPerm

The data in 'factpoison.csv' give survival times (in 10-hour units) from a 3 x 4 factorial experiment, the factors being (a) three poisons and (b) four treatments. Each combination of the two factors is used for four animals, the allocation to animals being completely randomized.

(a) Check ANOVA assumptions using residual vs fitted plots, a qq-plot and Levene's test.
The following code will help you to perform Levene's test on the data
leveneTest(Time ~ as.factor(Poison) * as.factor(Treatment), data = pos1a)

```
# 3.1a - Read in the data
pos1a <- read.csv("F:\\Lecture Courses\\STAT 293\\...\\ factpoison.csv")
str(pos1a)
```

```
## 'data.frame':   48 obs. of  3 variables:
## $ Poison   : int  1 1 1 1 1 1 1 1 1 1 ...
## $ Treatment: int  1 1 1 1 2 2 2 2 3 3 ...
## $ Time     : num  0.31 0.45 0.46 0.43 0.82 1.1 0.88 0.72 0.43 0.45 ...
```

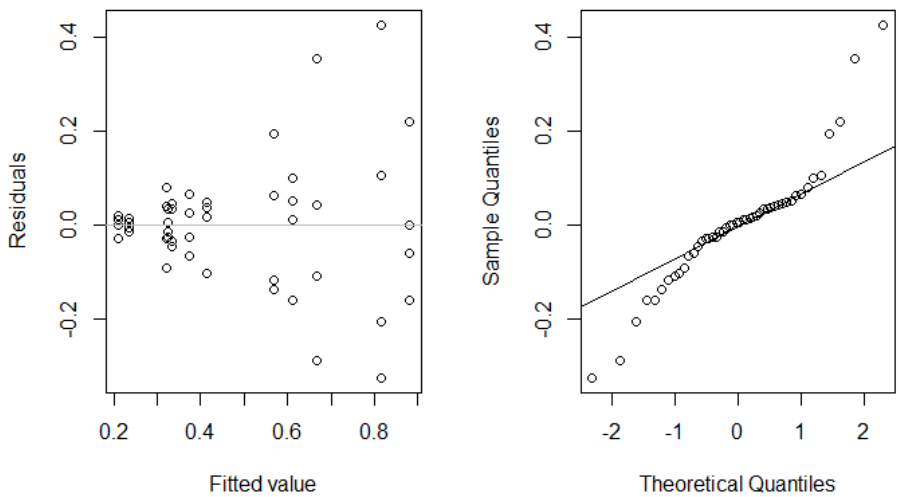
```
# 3.1b - Create the linear model
pos1b <- lm(Time ~  as.factor(Poison) + as.factor(Treatment) +

            as.factor(Poison) * as.factor(Treatment), data=pos1a)


# 3.1c - Obtaining a plot of residuals vs fitted values and QQ plot of residuals
par(mfrow=c(1,2))

plot(pos1b$res ~ pos1b$fit, xlab="Fitted value", ylab="Residuals")
abline(h=0,col="gray")

qqnorm(pos1b$res ,main="")
qqline(pos1b$res)
```



Residual vs Fitted Value Plot implies non-constant variance

qq-plot implies data is not normally distributed

```
# 3.1d - Conducting Levene's test for homogeneous variance
library(car)
leveneTest(Time ~ as.factor(Poison) * as.factor(Treatment), data = pos1a)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df        F value    Pr(>F)
## group 11        4.1323    0.0005833 ***
##        36

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Levene's test provides strong evidence against the null hypothesis that the data in each group have equal variance.

(b) Obtain an ANOVA table with p−values obtained using a normal theory ANOVA test, and an ANOVA table with p−values obtained from permutation tests. The aim of the experiment is to determine whether there is an effect of poison or treatment on survival times and whether there is any interaction between poison and treatment. For each of the two tests above give the hypotheses and state your conclusions. Comment on any similarities or differences in the results of the two tests.

```
# 3.1e - Obtain ANOVA table from Normal Theory ANOVA
anova(pos1b)
```

```
## Analysis of Variance Table

## Response: Time

 ##                                        Df        Sum Sq    Mean Sq    F value        Pr(>F)
## as.factor(Poison)                       2        1.03301   0.51651    23.2217        3.331e-07 ***
## as.factor(Treatment)                    3        0.92121   0.30707    13.8056        3.777e-06 ***
## as.factor(Poison):as.factor(Treatment)  6         0.25014  0.04169     1.8743        0.1123
## Residuals                          36 0.80073 0.02224

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
3.1f - Obtaining an ANOVA table from a permutation test
library(lmPerm)
set.seed(12345)

pos1c <- lmp(Time ~ as.factor(Poison) + as.factor(Treatment) +  as.factor(Poison) * as.factor(Treatment),
data=pos1a, perm="Prob")

anova(pos1c)
```

```
##Analysis of Variance Table
## Response: Time
```

	Df	R Sum Sq	R Mean Sq	Iter	Pr(Prob)
as.factor(Poison)	2	1.03301	0.51651	5000	<2e-16 ***
as.factor(Treatment)	3	0.92121	0.30707	5000	<2e-16 ***
as.factor(Poison):as.factor(Treatment)	6	0.25014	0.04169	2639	0.1201
Residuals	36	0.80073	0.02224		

Response Variable: Time
Factor A : Treatment
Factor A Levels: {1, 2, 3, 4}
Factor B : Poison
Factor B Levels: {1,2,3}

Normal Theory ANOVA	Permutation Test
$H_0: \ (\alpha\beta)_{ij} = 0 \quad \forall \ i,j$ $H_1: \ (\alpha\beta)_{ij} \neq 0 \ \ for\ some\ i,j$ p-value = 0.1123 => No evidence against the null hypothesis i.e. no evidence to suggest that there is an interaction between Poison & Treatment	*H_0 :* There is no interaction between Poison & Treatment *H_1 :* There is an interaction between Poison & Treatment p-value = 0.1201 => No evidence against the null hypothesis i.e. no evidence to suggest that there is an interaction between Poison & Treatment
$H_0: \ \alpha_i = 0 \quad \forall \ i = 1, \ \dots, 4$ $H_1: \ \alpha_i \neq 0 \quad for\ some\ i$ p-value = 3.777e-06 => Very strong evidence against the null hypothesis i.e. very strong evidence to suggest that Treatment has an effect on the response variable Time	*H_0 :* There are no differences in survival times among treatments *H_1 :* Survival times differ among treatments p-value = <2e-16 => Very strong evidence against the null hypothesis i.e. very strong evidence to suggest that Treatment has an effect on the response variable Time
$H_0: \ \beta_j = 0 \quad \forall \ j = 1, \ \dots, 3$ $H_1: \ \beta_j \neq 0 \quad for\ some\ j$ p-value = 3.331e-07 => Very strong evidence against the null hypothesis i.e. very strong evidence to suggest that Poison has an effect on the response variable Time	*H_0 :* There are no differences in survival times among Poisons *H_1 :* Survival times differ among Poisons p-value = <2e-16 => Very strong evidence against the null hypothesis i.e. very strong evidence to suggest that Poison has an effect on the response variable Time

In all cases we reach the same conclusion in the normal theory ANOVA and permutation tests. However, since the assumptions of normality and constant variance are not met, the p−values associated to the normal-theory ANOVA tests are not valid to draw conclusions from.